

Scavenging, Medical Gas Pipelines, and Vacuum Systems

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10.1 Introduction

Scavenging system is necessary to prevent contamination and pollution of the operating theater via collecting excess anesthetic gases and vapors from the breathing system and disposing them outside the working environment.

10.2 Definitions

Scavenging system is defined as a system for collecting and removing of vented anesthetic gases from the Operating Room (OR). In fact, these anesthetic gases and volatile anesthetics are exceeding the amount needed for the patient.

10.3 A Complete System Consists of Four Parts: (Fig. 10.1)

- 1. *Collecting tubes*: It is usually connected to APL valve and relief valve to collect any extra anesthetic gases and volatile anesthetics.

- 2. *Transferring tubes* (Fig. 10.2).
- 3. *Scavenging interface*: It is acts as a safety part and it protects the breathing circuit from any positive or negative pressure change.
- 4. *Disposal tubing*: It carries gas from interface to disposal assembly. *Gas disposal assembly*: which has two types, active or passive (old one), nowadays days active is the most common and it utilizes the hospital suction system.

10.4 Types of Scavenging Systems

10.4.1 Active Anesthesia Gas Scavenging Systems (AGSS)

There are several versions of the active Anesthesia Gas Scavenging System (AGSS) available depending on the hospital's type of waste gas disposal system. Each version has a 1-L reservoir to capture peak exhaust flows that briefly exceed the extract flow.



Fig. 10.1 Anesthesia gas scavenging system control panel

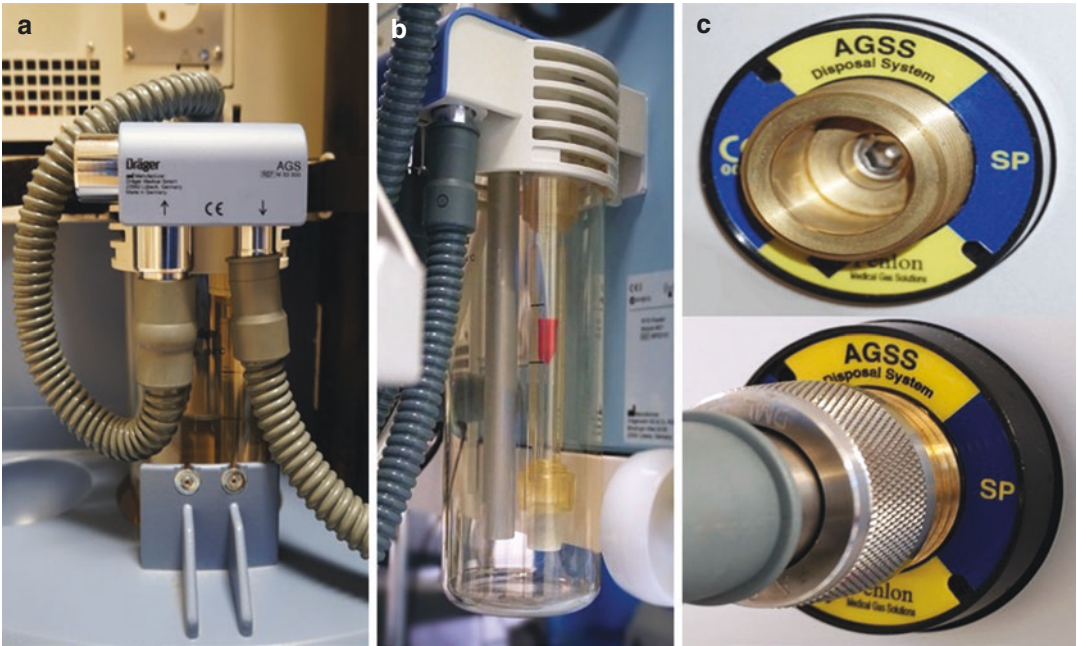


Fig. 10.2 (a, b) Scavenging system outlet at the back of anesthesia machine. (c) Scavenging system outlet

10.4.2 Active Adjustable Flow

It provides the capability to adjust the flow with a needle valve and a visual indicator bag which should be properly inflated. To ensure adequate scavenging, you have to adjust the needle valve. so, the visual indicator bag puffs out slightly with each breath. The bag should not be completely

collapsed (close the needle valve slightly) nor completely inflated (open the needle valve slightly) (Fig. 10.3).

Active Adjustable Flow gas scavenging can cause a high PEEP alarm. Make sure the scavenge hose is connected to suction. Make sure that the needle valve is open enough that the scavenger bag moves with inspiration and expiration.

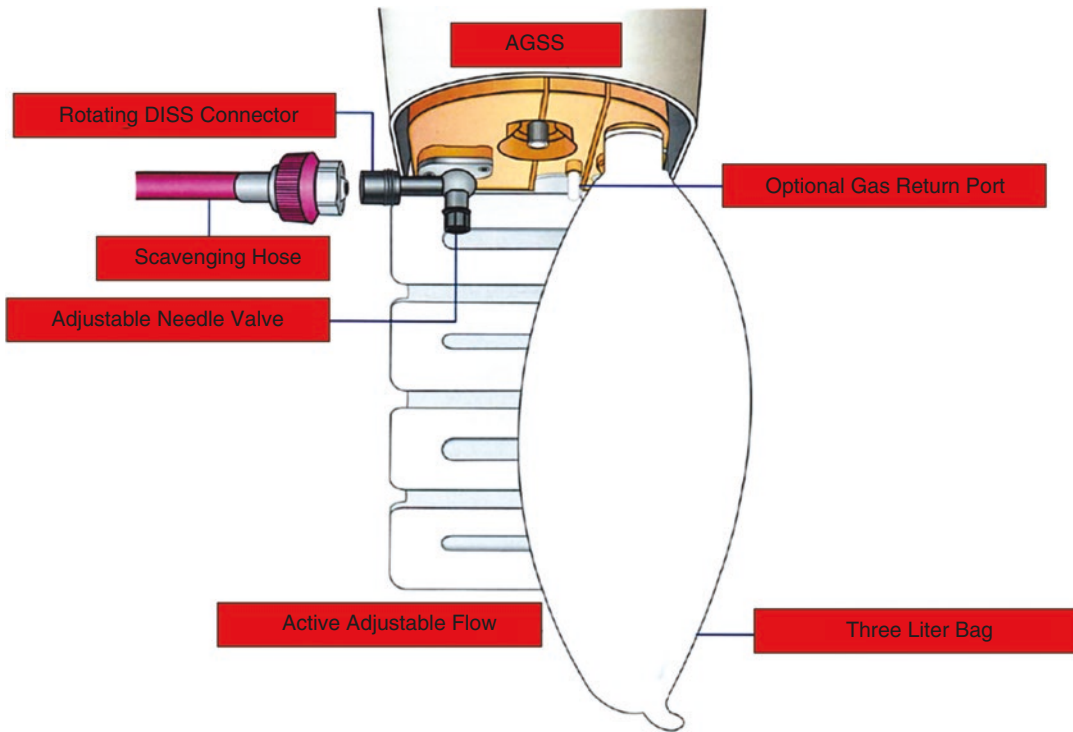


Fig. 10.3 Active anesthesia gas scavenging system (Active adjustable flow)

You can connect the sample gas exhaust tube to the gas return port. Exhaust gas will be directed to the scavenging system.

10.4.3 Passive AGSS (Anesthetic Gas Scavenging System) (Fig. 10.4)

Passive Anesthesia Gas Scavenging System (AGSS) is used for operating room environments that do not operate an active gas extraction system for waste gas disposal.

The passive AGSS contains both positive and negative pressure relief valves, to protect the breathing system.

Passive AGSS may be used with a non-recirculating ventilation system for waste gas disposal.

The tube connection from passive AGSS to the non-recirculating ventilation system should be an open connection, essentially at atmospheric pressure to an exhaust grill.

In passive AGSS, the waste gases proceed passively through the room ventilated exhaust gas of the OR.

Passive systems require that the patient must be protected from positive pressure buildup only.

10.5 Safety

10.5.1 Safety Valve for AGSS

Scavenging systems have safety features that prevent suction from being applied to the breathing circuit leading to negative pressure applied to the patient's lungs.

The anesthesia machines (Drager ZEUS), it has a valve to prevent anesthesia gas to return back to the machine, it is called (*Non-return valve to anesthesia gas scavenging system*) (Fig. 10.5).

Measuring levels of anesthetic agents in OR are important. The levels of both N_2O and halogenated agents in the OR can be measured by

Fig. 10.4 Passive anesthesia gas scavenging system

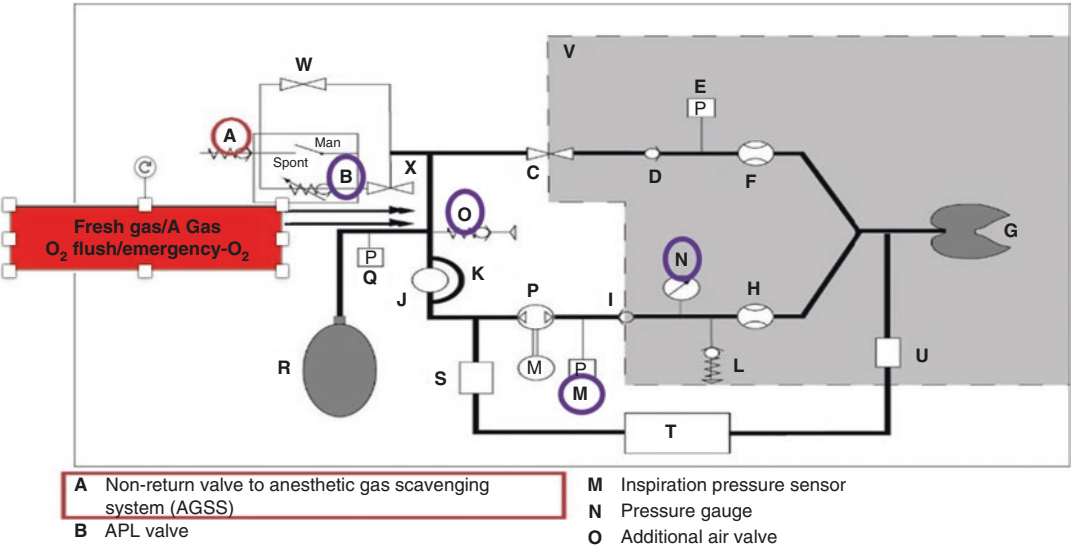
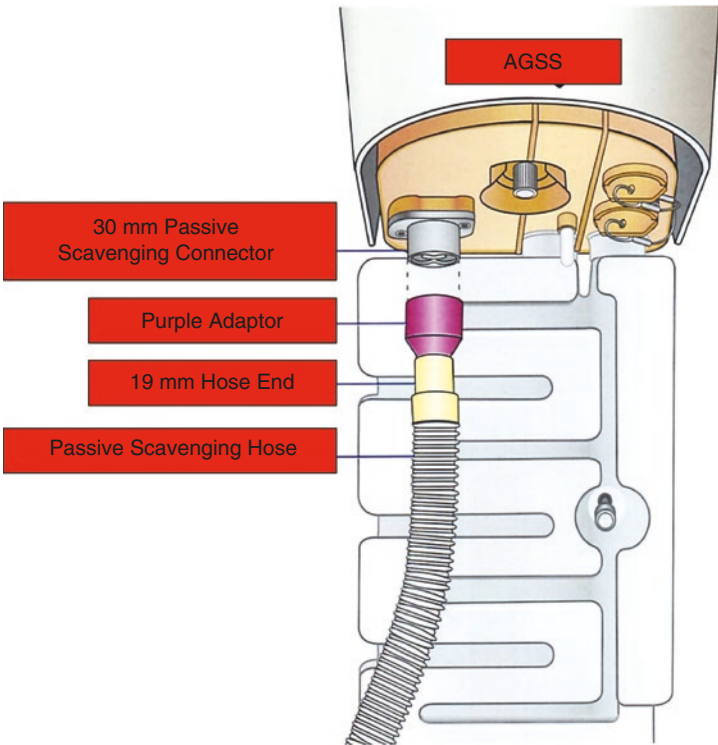


Fig. 10.5 Safety valve for AGSS

Infrared (IR) analyzers to trace levels of anesthetic agents.

10.5.2 Hazards of Scavenging

Scavenging of the anesthesia breathing circuit increases the complexity, and consequently the hazards, of administering anesthesia. If the scavenging interfaces were mistakenly bypassed or were to malfunction, excessive positive or negative pressure in the scavenging system could be directly transmitted to the breathing circuit.

This might cause cardiovascular embarrassment and barotrauma to the patient. Excessive negative pressure may be caused by unopposed vacuum and excessive positive pressure may be caused by occlusion of the connecting tubing.

In addition to occupational exposure for health care worker in close environment to anesthesia gas agents, barotrauma and ventilation dysfunction are a serious concern.

The effects of chronic exposure to volatile agents:

Some incidents have the potential to harm both patients and health care providers especially anesthesiologists, anesthesia providers, and PACU nurses are most directly affected by gases in the operating room and in the post-anesthesia care unit (PACU), where the patient continues to exhale physiologically gas in the surrounding atmosphere.

10.6 Avoiding Waste Gas Exposure

The following steps help to minimize gases exposure:

1. The scavenger system of the breathing system should be checked every day.
2. Tight mask fit is preferred during induction and recovery.
3. Prevent flow from breathing system into room air (while disconnecting or position the patient).

4. Avoid to washout anesthetics (into the breathing circuit) at the end of the anesthetic.
5. Do not spill liquid anesthetic agent.
6. Tight the vaporizers after filling.
7. Use low flows ventilation techniques.
8. Use cuffed tracheal tubes to minimize the leak.
9. Prevent leakage from Supraglottic Airway Devices (SAGD).
10. Check the machine regularly for leaks.
11. Disconnect nitrous oxide pipeline connection at wall at the end of the day.
12. Use total intravenous anesthesia (TIVA) if possible.

10.7 Scavenging System Troubleshooting Problems

10.7.1 High Pressures Scavenging System

It may result from a malfunction of pressure relief valve, the high pressures translate to increased circuit pressure and increased airway pressures, barotrauma may occur to the patient.

The patient has to be disconnected from the breathing circuit and ventilated by an alternative ventilation method. If the problem cannot be immediately corrected, a new anesthesia machine will need to be used.

10.7.2 Increased Circuit Pressures

- Check for the tube kinks.
- Check the vacuum control valve for proper adjustable (if it is there).
- Check the positive pressure release valve for proper functioning.

10.7.3 Decreased Circuit Pressures

- Check that the vacuum control valve if it is there for proper adjustable.
- Check the ventilator relief valve.

10.7.4 Overwhelmed Scavenging System

- Make sure the fresh gas flow is less than 10 L/min.
- Check the vacuum control valve if it there (you may need to increase it).
- Ensure the gas disposal conduit is functioning.

10.8 Recommendations

- The scavenger system of the breathing system should be checked every day.
- Adequate precautions should be taken to ensure the safety supply of medical gases.
- Anesthesia providers must be aware of the sources of medical gas to ensure safety and an adequate supply when delivering to patients.
- All hospitals should have a documented maintenance schedule for the ventilation system in

the Operating Room (OR), post-anesthesia care units (PACU), and the anesthesia machines.

10.9 Medical Gases

10.9.1 Introduction to Gas Cylinders and Medical Gases

Medical gases are used in all areas of a hospital. It constitutes an essential part of the operating theater, intensive care units, and for therapeutic treatment.

The most common medical gases are oxygen, nitrous oxide, and air, these gases are supplied via a large central source. Alternatively, they may be supplied via gas cylinders “E” size cylinders mounted on back of the anesthesia machine.

A waste anesthetic gas (WAG) scavenging system, medical suction system for surgical and anesthetic gases are provided centrally (Fig. 10.6).

Fig. 10.6 Central gases control panel HMC-HGH



10.9.2 Medical Gas System Components

Each medical gas is supplied by a separate system. It is essential that all components of each system are gas specific to ensure that there is no possibility of cross-connection between systems. Indeed, a common configuration is designed to each system including the following components.

1. Sources.
2. Piping network.
3. Valves.
4. Warning and alarm systems.
5. Outlets and inlets.
6. Secondary equipment.

10.9.3 Medical Gas Supply and Storage

10.9.3.1 Introduction

Medical gases are stored in purpose-built cylinders, ranging in capacity from 1.2 to 7900 L. Cylinders are fitted with different types of valves and cylinders are made of steel alloys or aluminum. Each medical gas cylinder is tested by visual inspection and with a hydraulic stretch test before being used to ensure it is safe to use.

Once tested, each medical gas cylinder is permanently stamped with a symbol that indicates its contents, service pressure, manufacturer's symbol, serial number, owner's symbol, test date, and testing facility.

After a cylinder has been tested and filled, a cylinder label and a separate batch label will be affixed indicating the contents of the cylinder, directions for use, batch number, fill, and expiration date. This information is very important if the cylinder involved to be in a recall or accident.

Figure 10.7 summarizes the color codes, physical state in cylinders, and pressure for different medical gases. Cylinders contents are under very high pressures, and safety precautions should

always be taken. Extreme temperatures and rapid temperature changes must also be avoided to prevent cylinder damage and/or leakage.

10.9.3.2 Valves

Common valves types include pin index valves (Fig. 10.2). The valve types on any given cylinder will typically vary according to the size.

Smaller cylinders (Type E cylinders), which are commonly used as backup on an anesthesia machine, usually have pin-indexed valves (Fig. 10.2).

Large cylinders, typically used to supply hospital pipelines, are usually fitted with bullnose valves.

The bullnose valve includes a non-interchangeable screw thread system, with a different number of threads per inch for each medical gas.

10.9.3.3 Cylinders Keys: (Opening/Closing)

Small cylinders (Type E) usually have a spindle, which may be opened or closed with a wrench. The correct sized wrench is usually permanently attached to the back of the anesthesia machine to prevent removal.

Larger cylinders (Type H) include a hand-wheel valve, which may be used to open or close the valves without any additional equipment.

10.9.3.4 Filling of Cylinder

Medical gas cylinders should not be overfilled or over pressurized because of the risk of accidental explosion or damage. It is important to keep in mind that the pressure inside a cylinder will vary with the ambient temperature

Each cylinder will have a service pressure stamped on it. To prevent damage or possible explosion, cylinders should never be filled above the service pressure.

For cylinders that contain liquefied gases, the filling limit is based on the filling density or percent ratio of the weight of a gas in the cylinder to the weight of the water in the cylinder. For both nitrous oxide and carbon dioxide, the filling density is 68%.

Gas	International Cylinder color	PSI at 21 Celsius	Physical state in cylinder	E-Cylinder capacity (liters)
Oxygen (O ₂)	White	1900-2200	Gas	660
Carbon dioxide (CO ₂)	Gray	838	Gas and liquid below 31°	1590
Nitrous oxide (N ₂ O)	Blue	745	Gas and liquid below 37°	1600
Air	White and black	1800	Gas	600

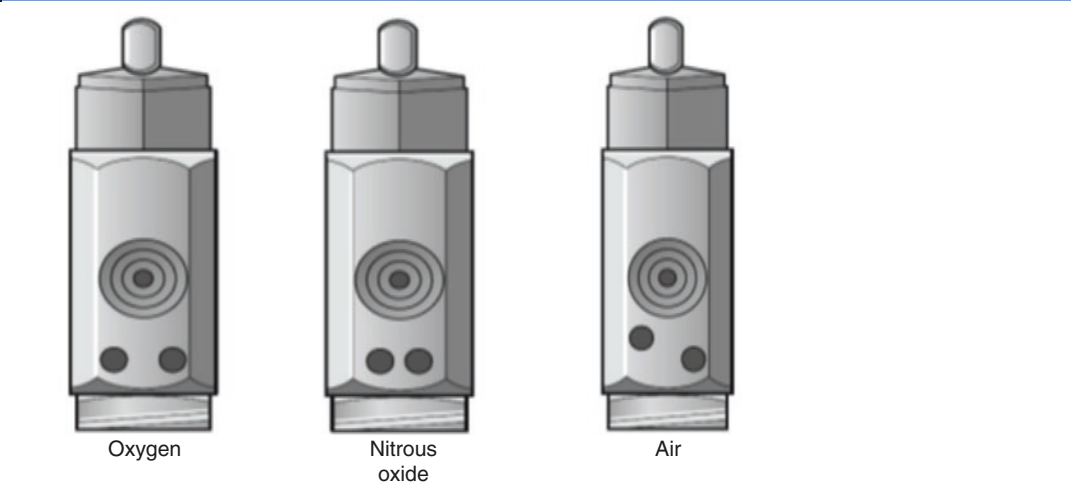


Fig. 10.7 PIN index system and physical properties for medical gas cylinders

Table 10.1 Types of medical gas cylinders

Gas	E-cylinder capacity ^a (L)	H-cylinder capacity ^a (L)	Pressure ^a (psig at 20°)	Color (USA)	Color (International)	Form
O ₂	625–700	6000–8000	1800–2200	Green	White	Gas
Air	625–700	6000–8000	1800–2200	Yellow	White and black	Gas
N ₂ O	1590	15,900	745	Blue	Blue	Liquid
N ₂	625–700	6000–8000	1800–2200	Black	Black	Gas

^aDepending on the manufacturer

Types of Cylinders (Table 10.1)

Gas cylinders are usually made of light weight chrome molybdenum steel, aluminum or special alloy such as aluminum alloy used in MRI.

There are different types of gas cylinders available to use, however, the most common

types used in anesthesia machine is E-type cylinder. These types have unique physical prosperities depending on the gas content, size, pressure limit, and volume limit.

Usually the E-Cylinder which used frequently in anesthesia machine is color coded according to gas content.

10.9.3.5 Safety Pressure Release Valves

All medical gas cylinders are fitted with a safety pressure release valve, which allows the gas inside to escape.

Types of safety release valves commonly used;

- Spring loaded pressure relief valves,
- Discs that rupture at a predefined pressure,
- Metallic plugs that melt at high temperatures.

Note: on cylinders with pin-indexed valves, the safety pressure release is located directly below the conical depression for the screw clamp. Care should, therefore, be taken not to damage this valve while mounting cylinders with pin-indexed valves onto the back of the anesthesia machine.

10.9.3.6 Medical Gas Cylinder Safety

- Medical gas cylinders should be properly stored and secured all the time to prevent damage or injury.
- Cylinders should never be dropped because a cracked pressurized cylinder can turn into a high-speed projectile missile.
- All cylinders should be stored away from open flame and in a dry, cool environment.
- All health care workers who are expected to handle medical gas cylinders should receive training and education about their proper use and storage.
- Medical gas cylinders should be always opened slowly to prevent rapid temperature rise from adiabatic expansion.
- Medical gas cylinders should be always kept closed when not in use.

Note

The medical gas cylinders are one of the simplest equipment used in the operating room, but it has the potential to be one of the most dangerous if not used and handled properly.

10.9.3.7 Calculating a Cylinder's Contents

It is possible to determine the contents of a medical gas cylinder that contains a compressed gas

(such as oxygen or air) using Boyle's law ($P_1V_1 = P_2V_2$) where P_1 stands for the atmospheric pressure, V_1 the volume of gas at atmospheric pressure, P_2 the pressure in the cylinder, and V_2 the water capacity of the gas in compressed state. This is particularly important in remote location, emergencies, and during transport.

Example

If you have an O_2 cylinder with pressure meter pointing 550 PSI and you are about to transfer a patient who need 3 L/min oxygen supply in order to keep his saturation above 92%.

How many minutes this oxygen cylinder can provide oxygen with the above-mentioned parameters? the answer:

The maximum pressure for oxygen cylinder is 2200 psi which contains at this pressure 700 L of O_2 .

To estimate how many liters, in O_2 cylinder we will use Boyle's law ($P_1V_1 = P_2V_2$)
 $v_2 = 550 \times 700 / 2200 = 175 \text{ L}$.

Now according to our patient, he is consuming 3 L/min of O_2 consequently the available O_2 in the cylinder (175 L) will be provided for 58.3 min (175/3).

Note that there is simple equation available and online calculator to help with calculation which are useful in daily practice (<http://www-users.med.cornell.edu/~spon/picu/calc/o2tankd.htm>).

10.9.3.8 Medical Gas Pipeline Network and Manifold

The main oxygen supply in the operating room is the hospital pipeline. This system delivers oxygen at 55 pounds per square inch gauge (PSI) and comes from one of two sources:

1. A primary liquid oxygen tank (with either a manifold of compressed gas cylinders or a smaller secondary liquid oxygen tank as a backup).
2. Two banks of compressed gas cylinders (with a smaller bank of compressed gas cylinders as a backup).

All hospitals should always have at least a 2-day supply of oxygen on-hand, and a backup

supply with at least a 1-day supply. (The total amount required will depend on the hospital capacity and specific needs).

The high-pressure oxygen source connects to the hospital pipeline through a two-stage pressure regulator. In hospitals where a manifold of oxygen cylinders is used, only one of the two oxygen banks will supply the main pipeline at any time. Once the first bank exhausted, the second bank will automatically switch over. This allows the finished cylinders to be changed out, without having to disrupt the whole system. The entire system is typically monitored by a centralized control station where visual indicators show the status of the two banks at all times.

10.9.3.9 Safety of Medical Gas Pipeline Network (Fig. 10.8)

All of the main hospital gas supplies and manifolds are typically physically located outside of the hospital. A network of supply pipelines conveys gas from the manifold to the required delivery points around the hospital campus. Safety standards for both oxygen and other positive pressure medical gases require the use of copper tubing to prevent the spontaneous combustion of organic oils. The supply networks are designed with pressure monitors and shut off valves throughout the system to allow isolation of specific areas for maintenance or emergency repairs.

Supply lines typically have the contents and flow direction labeled at regular intervals.

10.9.3.10 Medical Gas Outlets

All hospital pipelines will end in terminal wall outlets. These color-coded outlets come in one of two varieties: (1) Diameter index safety system (DISS) outlets or (2) non-interchangeable quick coupling connectors.

DISS The DISS was developed to prevent accidental misconnections of different medical gases.

The connector system makes it physically impossible to connect the wrong hose to the wrong pipeline. Each connector is made up of a body, nipple, and nut. The body has two concentric bores, which match specific shoulders on the matching nipple. The diameters of the bores are different for each gas, making them non-interchangeable. Only appropriately matched parts will fit together and allow a complete connection.

Non-interchangeable Quick Connectors

Quick connectors allow flow meters, hoses, machines, and other pieces of equipment to be quickly connected/disconnected without using any tools or undue force. Each quick connector is made up of a pair of gas-specific male and female

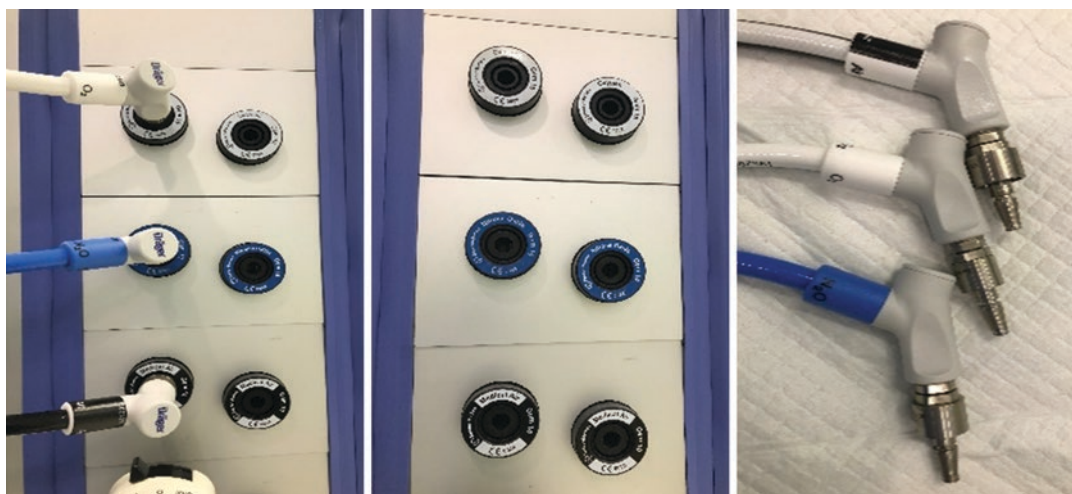


Fig. 10.8 (a) Medical gas outlets, (b) wall outlet sockets (c) Medical gas pipelines

parts. The two components are then locked together by a releasable spring mechanism. Different design prevents hoses from being placed into the wrong outlet. While quick connectors may be easier to use than DISS connections, however, they tend to leak with a higher frequency.

10.9.3.11 Pipeline Alarm Management System

The Gas Management System is a key component of every hospital. It is designed to provide an early warning indication for abnormality and to ensure optimum status of the central gas supply at all time.

10.9.4 Oxygen

Oxygen is an essential drug used by every anesthesiologist. Oxygen supply begins with the liquefaction of compressed air. Oxygen is separated from liquid air by taking advantage of the differences in the boiling points of oxygen and nitrogen. This enables evaporation of nitrogen first leaving liquid oxygen behind, which is later on evaporated and collected.

In the medical environment, oxygen can be supplied as either compressed gas from room temperature cylinders or as liquid oxygen from a cryogenic liquid system container. The system required to supply and store liquid oxygen is more expensive than regular oxygen cylinders; however, it is often more economical in facilities in which higher volumes of oxygen are used. This is because liquid storage is less bulky and less costly than the equivalent capacity of high-pressure gaseous storage.

10.9.4.1 Liquid Oxygen Storage Systems

The liquid oxygen storage system consists of one or more cryogenic storage tanks, one or more vaporizers, a pressure control system, and the piping necessary to support all of the requisite fill, vaporization, and supply functions. Liquid oxygen must remain under pressure and below its critical temperature of -118°F to remain as a liquid. Because the temperature gradient between

liquid oxygen and the surrounding environment is significant, keeping liquid oxygen well insulated from the surrounding heat is critically important.

Vaporizers attached to the system convert liquid oxygen into a gaseous state. Downstream, a pressure control manifold adjusts the gas pressure that is provided to the outgoing pipelines. Most cryogenic liquid systems include a backup system, which typically consists of another smaller sized liquid oxygen container or a separate manifold of oxygen cylinders. In the event of oxygen supply failure, anesthesia providers are called upon to ensuring patient safety.

10.9.4.2 Oxygen Concentrators

An oxygen concentrator provide oxygen from atmospheric air. It is used as the primary oxygen source in some remote locations. These devices work through the adsorption of atmospheric nitrogen by a molecule sieve. To ensure accurate delivery of oxygen, the oxygen output concentration should be carefully monitored when these devices are used.

10.9.4.3 The Dangers of Liquid Oxygen

As liquid oxygen is stored in medical gas cylinders at -118°C , it can cause immediate damage of tissue on

Contact.

10.9.4.4 Compressed Air System

Compressed air systems provide a continuous supply of medical compressed air for ventilation purposes and for operating surgical instruments.

10.9.4.5 Recommendation

- The connections between the anesthesia machine and the wall supply should be checked daily to ensure there is no physical damage or misconnection.
- The pressures in the anesthesia machine should be checked prior to each theater list to ensure that there is an adequate quantity of medical gas available.
- Regular maintenance of pipeline, machines, and alarms.

- Make sure you know what is your hospital policy for oxygen supply failure and/or similar emergencies.

10.10 CO₂ Gas Analyzer

10.10.1 Gas Analyzers Troubleshooting Problems

CO₂ tracing wave appears on the machine does not appear.

Ensuring a proper CO₂ tracing is part of the machine check out. We should identify any machine defect early but if no tracing appears during a procedure, we should check the following:

- First confirm that it is not a lack of patient ventilation.
- Check if the anesthesia machine was calibrated.
- Check if gas analyzer has been turned on and warm-up has completed.
- Check if the gas analyzer is not performing a temporary calibration.
- Check for sample line disconnect or obstruction.
- Anesthesia provider should confirm ETT placement.
- Check for a circuit disconnect.
- Check that the water trap is inserted properly and not full of water.
- Check for any recent aerosolized medication administration, when aerosolized medication is given through the ETT, the gas analyzer may have trouble identifying the various gases.

The common cause for inaccurate readings from the gas analyzer is condensation in the sample line or an incomplete connection to the machine (Kinked, fractured) in addition poorly connected sample lines can also cause faulty readings.

10.10.2 Troubleshooting Common Problems with CO₂

Elevated CO₂ Wave on Capnography

- Examine inspiratory and expiratory flow valves.
- Check if an exhausted CO₂ absorber because commonly high inspired CO₂ is due to exhausted absorber and replace the CO₂ absorber.
- Faulty inspiratory or expiratory flow valve can increase CO₂, the valves can become incompetent if they are damaged, dried out, cracked, or get stuck.

10.11 Vacuum (Suction System)

Generally, a vacuum is a volume of space that has no matter in it. It results in a space having a much lower (negative) pressure than atmospheric pressure. Vacuums are essential in medical practice because they are used to create suction. Suction can then be used within the operating room to remove unwanted gases, liquids, and solid materials from both patient and the environment. Hospitals typically provide a vacuum to each patient care location via a pipeline system, which is capable of delivering a vacuum of close to 300 mm Hg at each terminus. Vacuum pipelines are typically constructed the same way medical gas supply lines are (copper tubing) but are usually larger in diameter. A vacuum is created placing two pumps in parallel with a reservoir between them to (1) even out the vacuum and (2) remove any debris from the system. Filters, before and after the wall connection, are important to prevent major leakage of harmful or contaminated materials into the central vacuum system.

In most modern hospitals, vacuum is available from wall outlets located throughout the building.

10.12 Maintenance of Suction Equipment

1. To ensure optimum performance of suction equipment, regular maintenance and inspection are important. Every hospital should have a maintenance program to minimize costly breakdowns and keep equipment operating at peak efficiency. Here is a suggested to make checklist for care of suction equipment after each patient use:
2. Thorough cleaning of reusable equipment and disposal of single-use components.
3. Careful inspection of the equipment, with special attention to filters.
4. Analysis of performance.
5. Adjustment and repair if necessary.
6. Re-cleaning.
7. Sterilization, if appropriate.
8. *biohazardous materials should be disposed according to local hospital guidelines.*
9. *any disposable parts of the suction system should be disposed properly and never ever to reused again.*
10. *personal protective equipment should be wearing during suction cleaning with contaminated materials.*
11. *Gloves, Face and Eye Protection.*
12. *Before start cleaning any electrical connection to suction system should be disconnected.*

Take-Home Message

Medical gases are a critical element of the practice of anesthesiology, but also a potential source of harm for patients and anesthesia providers alike. It is, therefore, critical that personnel who work in the operating room understand how to properly use, maintain, and store medical gases.

Although the practice of anesthesiology is constantly changing, suctioning is an important part of daily patient care and in many cases can be a potentially life-saving procedure. A thorough understanding of the physics involved in creating suction, the factors affecting flow rate, and the clinical applications of negative pressure will make the procedure more effective and as safe as possible.

Scavenging system (waste gas) dispose all ventilated gases emitting from the patient and anesthesia machine.

It has a crucial rule in the operating room environment to prevent any pollution which could be hazardous.

The National Institute for Occupational Safety and Health (NIOSH) recommends limiting the room concentration of nitrous oxide to 25 ppm and halogenated agents to 2 ppm.

Scavenging system is integral part of anesthesia machine that it has to be familiar to anesthesiologists and anesthesia technicians. Also it should be monitored in between the cases frequently as well.

American Society of Anesthesia stressed that it is important by being a part of daily anesthesia machine checklist.

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